Python Beginner Practice Set

10 Basic Projects

1. Calculator

Question:

Write a program that takes two numbers and an operator (+, -, *, /) as input from the user and prints the result.

Code Answer:

```
num1 = float(input("Enter first number: "))
operator = input("Enter operator (+, -, *, /): ")
num2 = float(input("Enter second number: "))

if operator == "+":
    result = num1 + num2
elif operator == "-":
    result = num1 - num2
elif operator == "*":
    result = num1 * num2
elif operator == "/":
    if num2 == 0:
        result = "Error: Division by zero"
    else:
        result = num1 / num2
else:
    result = "Error: Invalid operator"

print("Result:", result)
```

2. Number Guessing Game

Question:

Write a program where the computer randomly selects a number between 1 and 100, and the user has to guess it. The program should tell the user if their guess is too high, too low, or correct, and keep asking until the correct number is guessed.

Code Answer:

```
import random

secret_number = random.randint(1, 100)
guess = None

while guess != secret_number:
    guess = int(input("Guess a number between 1 and 100: "))

    if guess < secret_number:
        print("Too low! Try again.")
    elif guess > secret_number:
        print("Too high! Try again.")
    else:
```

```
        print("Correct! You guessed the number.")
```

3. To-Do List

Question:

Write a program that lets the user add tasks, view all tasks, and remove a task from a to-do list using a menu, until the user chooses to exit.

Code Answer:

```
todo_list = []

while True:
    print("\n1. Add Task")
    print("2. View Tasks")
    print("3. Remove Task")
    print("4. Exit")
    choice = input("Enter your choice: ")

    if choice == "1":
        task = input("Enter task: ")
        todo_list.append(task)
        print("Task added.")
    elif choice == "2":
        if len(todo_list) == 0:
            print("No tasks in the list.")
        else:
            print("Your Tasks:")
            for index, task in enumerate(todo_list, start=1):
                print(f"{index}. {task}")
    elif choice == "3":
        task_number = int(input("Enter task number to remove: "))
        if 1 <= task_number <= len(todo_list):
            removed_task = todo_list.pop(task_number - 1)
            print(f"Removed task: {removed_task}")
        else:
            print("Invalid task number.")
    elif choice == "4":
        print("Exiting To-Do List. Goodbye!")
        break
    else:
        print("Invalid choice. Try again.")
```

4. Password Generator

Question:

Write a program that generates a random password of a length specified by the user, using a mix of letters, digits, and special characters.

Code Answer:

```
import random
import string
```

```
password_length = int(input("Enter desired password length: "))

characters = string.ascii_letters + string.digits + string.punctuation
password = "".join(random.choice(characters) for i in range(password_length))

print("Generated Password:", password)
```

5. Simple Quiz

Question:

Write a program that asks the user 3 multiple-choice questions, checks their answers, and displays the final score out of 3 at the end.

Code Answer:

```
score = 0

print("Question 1: What is the capital of France?")
print("a) Berlin  b) Madrid  c) Paris  d) Rome")
answer1 = input("Your answer: ").lower()
if answer1 == "c":
    score = score + 1

print("\nQuestion 2: Which language is used for web styling?")
print("a) HTML  b) CSS  c) Python  d) Java")
answer2 = input("Your answer: ").lower()
if answer2 == "b":
    score = score + 1

print("\nQuestion 3: What is 5 + 7?")
print("a) 10  b) 11  c) 12  d) 13")
answer3 = input("Your answer: ").lower()
if answer3 == "c":
    score = score + 1

print(f"\nYour final score is {score} out of 3.")
```

6. Unit Converter

Question:

Write a program that converts a distance value entered by the user from kilometers to miles.

Code Answer:

```
kilometers = float(input("Enter distance in kilometers: "))
miles = kilometers * 0.621371

print(f"{kilometers} kilometers is equal to {miles:.2f} miles")
```

7. Even/Odd Checker

Question:

Write a program that takes a number as input from the user and checks whether it is even or odd.

Code Answer:

```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print(f"{number} is Even")
else:
    print(f"{number} is Odd")
```

8. Temperature Converter

Question:

Write a program that converts a temperature value from Celsius to Fahrenheit based on user input.

Code Answer:

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9/5) + 32

print(f"{celsius}°C is equal to {fahrenheit}°F")
```

9. Simple Interest Calculator

Question:

Write a program that calculates simple interest based on principal amount, rate of interest, and time period entered by the user.

Code Answer:

```
principal = float(input("Enter principal amount: "))
rate = float(input("Enter rate of interest (in %): "))
time = float(input("Enter time period (in years): "))

simple_interest = (principal * rate * time) / 100

print(f"Simple Interest: {simple_interest}")
```

10. Rock Paper Scissors

Question:

Write a program where the user plays Rock, Paper, Scissors against the computer. The computer should choose randomly, and the program should declare the winner of each round.

Code Answer:

```
import random

choices = ["rock", "paper", "scissors"]

user_choice = input("Enter rock, paper, or scissors: ").lower()
```

```
computer_choice = random.choice(choices)

print("Computer chose:", computer_choice)

if user_choice not in choices:
    print("Invalid choice. Please try again.")
elif user_choice == computer_choice:
    print("It's a tie!")
elif (user_choice == "rock" and computer_choice == "scissors") or \
     (user_choice == "paper" and computer_choice == "rock") or \
     (user_choice == "scissors" and computer_choice == "paper"):
    print("You win!")
else:
    print("Computer wins!")
```